How-To Library

Informatica

Import a Data Domain Profile from Informatica Developer into Test Data Management

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Abstract

You can import profiles that you create in Informatica Developer into Test Data Manager. This article documents the steps to create a data domain profile in Informatica Developer and import the profile into Test Data Manager.

This article assumes that you have knowledge of Informatica Developer and Test Data Management (TDM) and are familiar with operations in Informatica Developer and Test Data Manager.

Supported Versions

- Test Data Management 9.7.1 HotFix 1
- Test Data Management 9.7.1

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Overview

You might want to analyze data from a data source that you cannot perform data discovery on in Test Data Manager. You can import profiles from the Model repository and view the profiles in Test Data Manager.

For example, you might have flat files that contain data. You cannot run profiles on flat file sources in Test Data Manager. You can run the profile in Informatica Developer and import it into Test Data Manager. You can then apply the results of the profile and perform other TDM operations such as data masking or data subset on the source data.

To view a profile that you import, you must import the source metadata and profile objects such as data domains into TDM.

Example Scenario

An organization provides application testing services to customers. Each customer provides some production data or sample data to use in different test cases.

The organization uses TDM to create test data from the production or sample data. The test team performs data discovery to identify sensitive data before it uses the data to create test data. Some customers provide the data in the form of flat files. You cannot perform data discovery on flat files in Test Data Manager. You can import the profile from the Model repository and view it in Test Data Manager.

The organization uses Informatica Developer to run a profile on the flat file data and then imports the profile into Test Data Manager. It can then analyze the results in Test Data Manager and perform other TDM tasks on the data as required.

To run a profile in Informatica Developer and import it into Test Data Manager, perform the following tasks:

1. Create a mapplet with the required expression that you want to use in the data domain. Validate the mapplet as a rule.
2. Create a data domain and add the mapplet to the data domain.
3. Create and run an Enterprise Discovery profile to analyze the data based on the data domain that you created.
4. Import the data domain into Test Data Manager.
5. Create a project and import the source metadata from the PowerCenter repository into the project.
6. Import the profile into the project in Test Data Manager.

Prerequisites

Before you run a profile in Informatica Developer and import it into Test Data Manager, perform the following tasks:

- Ensure that the Informatica Developer version is compatible with the TDM version.
- Add the flat file sources to the source files location in the Informatica server. The default location is `<Informatica installation directory>\server\infa_shared\SrcFiles`.
- Create a folder in Repository Manager and add the flat file sources to the folder.
- Import the flat file sources to the required folder in Informatica Developer. Ensure that the run-time location points to the files in the Informatica server location.
- Import the column headers when you import the files into Informatica Developer. By default, Informatica Developer defines headers in Camel case and the PowerCenter client defines headers in uppercase. The mismatch in metadata might cause issues when you view the profile results. You might also run into this issue if the flat files do not contain headers and you import with the default options.

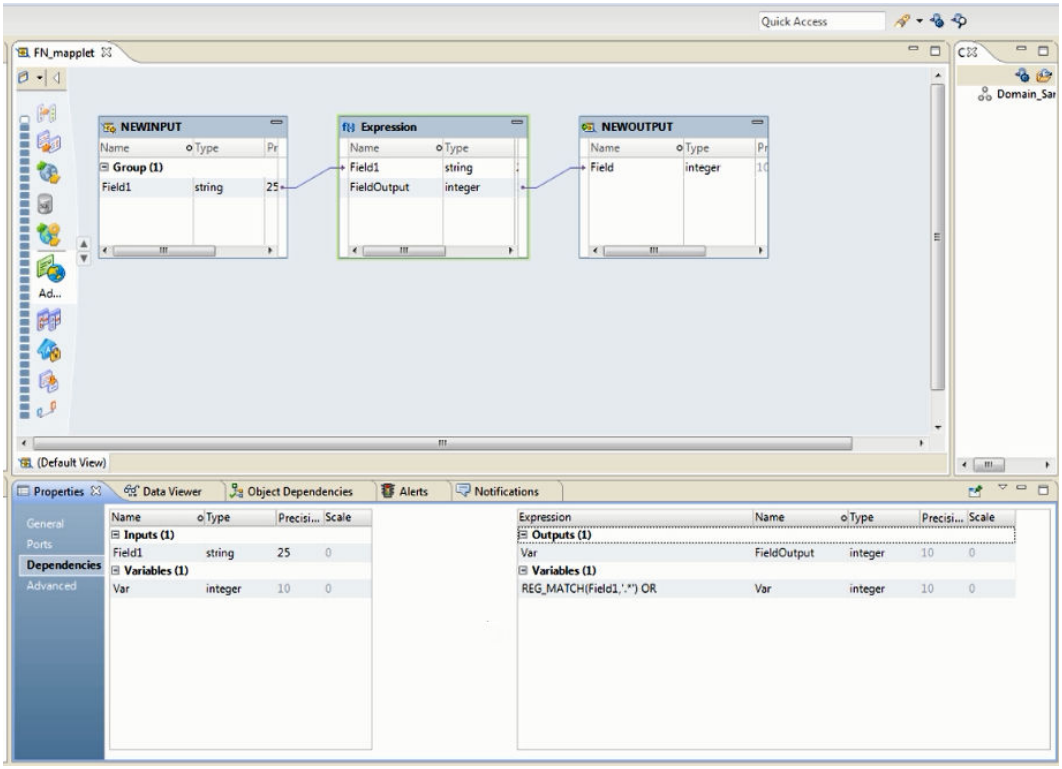
Step 1. Create a Mapplet in Informatica Developer

Create a mapplet that contains the regular expression against which you want to analyze the data.

1. Select the required folder in the **Object Explorer** view.
2. Click **File** > **New** > **Mapplet**.
3. Enter a mapplet name.
4. Click **Finish**.

   An empty mapplet appears in the editor.
5. Add an input, output, and expression transformation to the mapplet.
6. Enter the required regular expression that includes a data pattern and a metadata pattern in the expression transformation.

The following image shows the mapplet with the transformations:



7. Validate the expression to ensure it is correct and save the mapplet.
8. Validate the mapplet as a rule.

Step 2. Create a Data Domain in Informatica Developer

Create a data domain and add it to the data domain glossary. The data domain must use the mapplet that you created to analyze column names and column data.

1. Click **Window** > **Preferences**.

   The **Preferences** dialog box appears.
2. In the **Preferences** dialog box, select **Informatica** > **Data Domain Glossary**.

   The Developer tool displays a list of all data domains in the **Data Domain Glossary** panel.
3. In the **Data Domain Glossary** panel, select **Data Domains**.
4. Click **Add**.

   The **Data Domain** dialog box appears.
5. Enter a name and description.
6. Choose to use the data rule to discover data domains based on column data and column names in the data source.
7. Click **Browse** to open the **Select Location** dialog box.
8. Select the appropriate mapplet rule and click **OK**.

   When you create a data domain, the Developer tool links mapplets and other dependent objects associated with the data domain to the data domain glossary. The mapplet that you select appears in the data rule and column name rule fields.

9. Click **Finish**.

   The Developer tool adds the data domain to the data domain glossary.

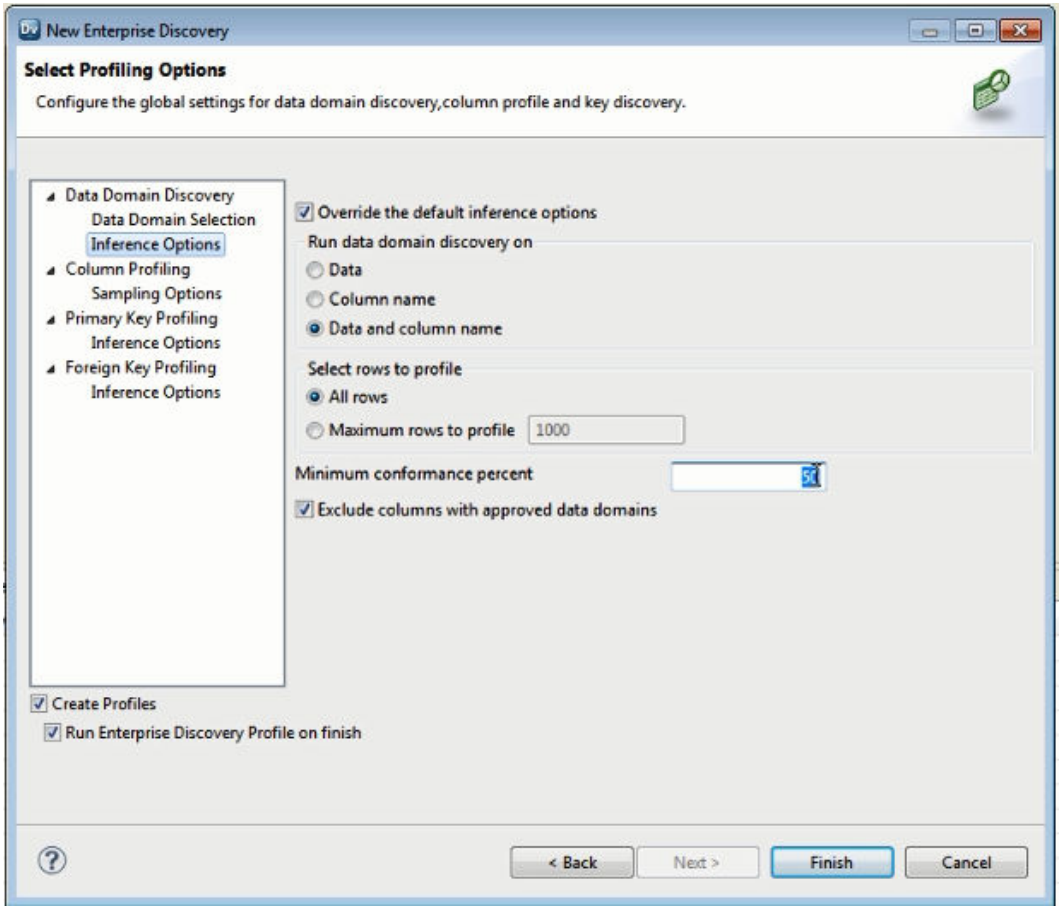
Step 3. Create and Run a Data Domain Profile in Informatica Developer

Run an Enterprise Discovery profile in Informatica Developer to import and view the profile in Test Data Manager.

1. In the **Object Explorer** view, select the folder that contains the data objects that you want to run a profile on.
2. To open the **New Profile** wizard, click **File** > **New** > **Profile**.
3. Select **Enterprise Discovery Profile** and click **Next**.
4. Enter a name. You can add a description.
5. Click **Next**.
6. In the **Add Data Objects to Profile Definition** window, click **Choose**.
7. Browse and select the required flat file data objects and click **OK**.

   The files appear in the list of data objects.
8. Click **Next**.
9. In the **Add Resources to Profile Definition** window, click **Next**.
10. In the **Select Profiling Options** window, select **Data Domain Selection** and select the required data domain.
11. Select **Inference Options** and choose to override the default inference options.
12. Choose to run data domain discovery on data and column names.
13. Enter the number of rows to run the profile on and a minimum conformance percent.

The following image shows the profiling options that you configure for the Enterprise Discovery profile:



14. Select **Column Profiling** > **Sampling Options** and disable the column profile.
15. Select **Primary Key Profiling** > **Inference Options** and disable the primary key profile.
16. Select **Foreign Key Profiling** > **Inference Options** and disable the foreign key profile.
17. Select **Create Profiles** and **Run Enterprise Discovery Profile on finish** and click **Finish**.
18. Refresh the repository connection when the profile completes.
19. In the **Object Explorer** view, select the **Profiles** folder within the folder that contains the data objects.
20. Open the profile and click **Data Domains** to view the results of the data domain profile.

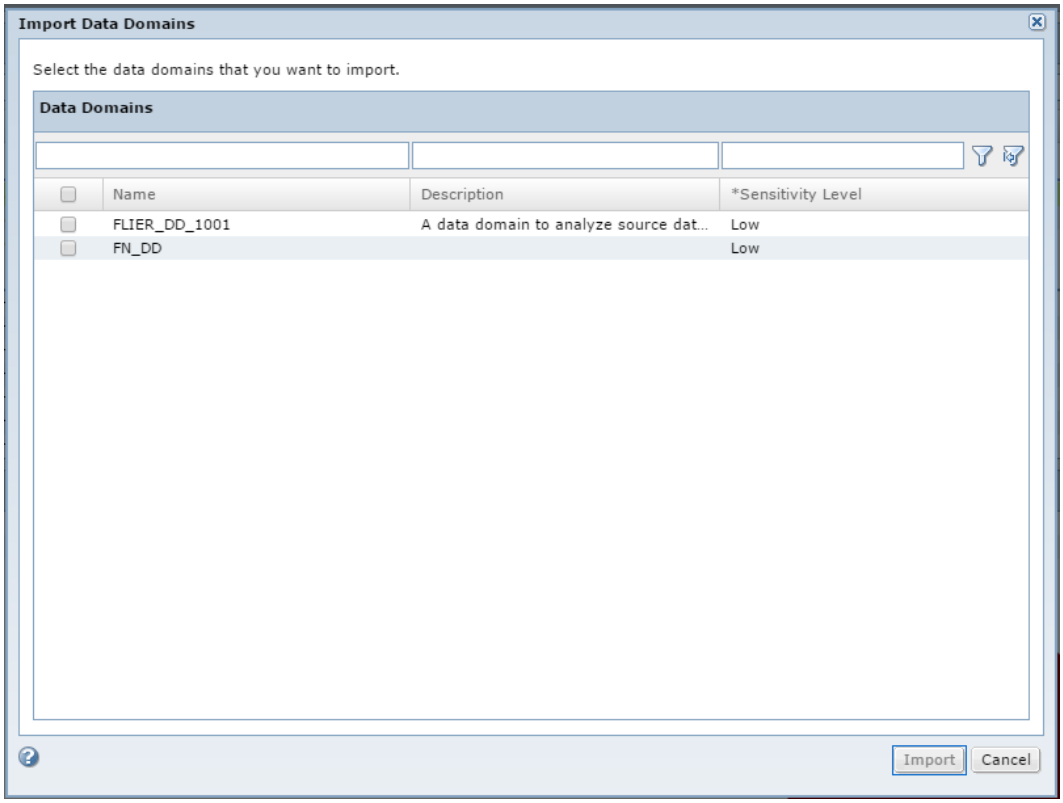
Step 4. Import the Data Domain into Test Data Manager

Before you import a data domain profile into Test Data Manager, you must import the data domains in the profile. You can import data domains from the Model repository into TDM.

1. Log in to Test Data Manager and click the **Policies** view.
2. Click **Actions** > **Import Data Domains**.

   The **Import Data Domains** window opens with a list of data domains in the Model repository.

The following image shows the **Import Data Domains** window:



3. Select the data domain that you created in Informatica Developer.
4. Assign a sensitivity level to the data domain. Click the **Sensitivity Level** column and choose the required level from the list. Click **Save**.
5. Click **Import**.

   If a data domain with the same name exists, you can choose to overwrite the data domain or cancel the import. TDM imports the data domain with the description and any links to business terms in the Business Glossary. You can view the data domain and the properties in the list of data domains.

Step 5. Create a Project and Import the Data Source

Create a project and import the source metadata. You cannot view the profile in a project if it does not contain the source metadata.

1. To access the projects, click **Projects**.
2. Click **Actions** > **New**.
3. In the **Create Project** dialog box, enter a name and an optional description for the project.
4. Click **OK**.

   The project opens in a new page.
5. Click **Actions** > **Import Metadata**.

   The **Import Metadata** window appears.
6. Choose to import metadata from the PowerCenter repository.
7. Click **Next**.

8. Select the PowerCenter folder that contains the flat file data sources that you want to import.
9. Click **Next**.
10. Select the files that you want to import.
11. Click **Next**.
12. Choose to import the sources immediately.
13. Click **Finish**.

    View the progress of the import job in the **Monitor** view. After the job finishes, access the imported metadata through the **Data Sources** details view.

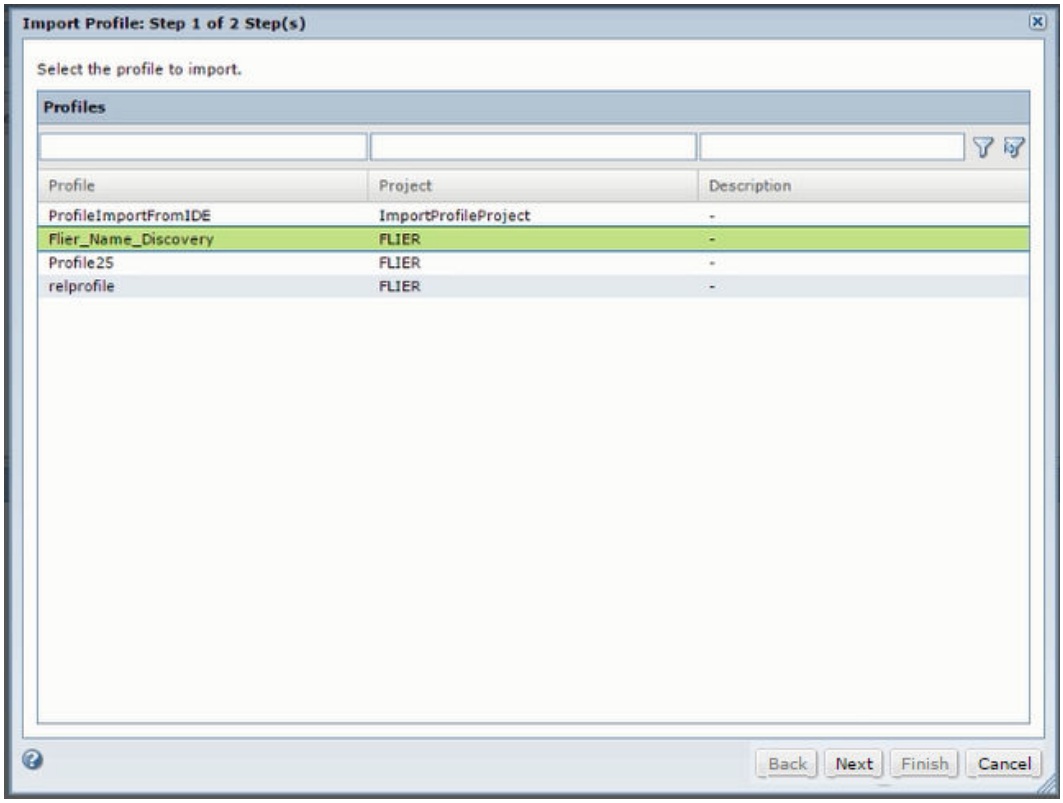
Step 6. Import the Profile from Informatica Developer

Import the profile that you created in Informatica Developer to view it in Test Data Manager.

1. Open the project and click the **Discover** | **Profile** view.
2. Click **Actions** > **Import Profile**.

   The **Import Profile** dialog box appears.
3. Select the profile that you want to import from Informatica Developer. Click **Next**.

   You cannot import a profile that contains a table name with special characters other than underscore (_). The following image shows the **Import Profile** window:



4. Optional. Edit the profile name and description if required. TDM imports the profile with the same name and description.
5. Verify that the correct profile type is selected. By default, TDM selects all the profile types that the profile contains. You can choose the profile types that you want to import.

6. Click **Finish**.

   The imported profile appears in the **Discover** view.

7. Click **Monitor** and verify that the import job is complete.
8. Open the **Discover** tab and click the **Refresh** button to verify that the status has changed to **Review Required**.

You can now review the results, approve or reject the data domain assignment for columns, and mark the data domain review as complete. You can then use the data domain assignment to perform TDM operations on the source data.

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